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Optimizing e-commerce rankings: insights and implications for Amazon sellers

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Abstract. Product ranking is critical to user engagement and sales performance on e-commerce marketplaces. Despite its significance, limited research focused on analyzing the features and temporal dynamics that influence product ranking from the seller's perspective. Using a dataset of *28,406 unique product listings scraped from Amazon.be* between March 21 and 27, 2025, this study investigates the key factors influencing product ranking and the stability of that ranking over time. Employing a *Gradient Boosted Tree Learning-to-Rank model (LambdaMART)*, the analysis reveals that sales rank is the most influential factor in determining product position, followed by price competitiveness and semantic similarity between title and search query. Additionally, *survival analysis* highlights that while sponsored products tend to exhibit greater short-term ranking stability, they are significantly more likely to drop to a lower rank over time compared to organic listings. These findings suggest that ad placements should be viewed as tactical tools for short-term visibility, whereas sustained success on Amazon requires strong organic performance, driven by factors such as competitive pricing, high-quality content, and reviews. As a result, this research provides actionable insights for sellers to strategically apply paid and organic approaches to optimize and maintain product visibility in Amazon's competitive marketplace.

Keywords: E-commerce product rank, Learning-to-Rank, LambdaMART, survival analysis, Amazon ranking, Search Engine Optmization

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Introduction

Online marketplaces have become essential platforms for businesses to sell products and for customers to find what they need quickly and efficiently [33]. These platforms bring together three primary stakeholders: the e-commerce marketplace itself, the sellers, and the customers. Customers use these platforms to reduce search costs when looking for products [63]. In response, e-commerce platforms develop and refine ranking algorithms to display the most relevant products at the top of search results [63]. The quality of the rankings greatly influences customer purchase decisions and, consequently, platform revenue [28]. Products that rank higher will gain more visibility, attract more clicks, and lead to increased sales. Therefore, both sellers and platforms are highly invested in improving product rankings.

Amazon stands out as the most influential marketplace [60]. As of 2022, it has 300 million monthly active users [57]. Research shows that 56% of customers begin their product search on Amazon, highlighting its dominance as a product search engine [30]. Amazon offers paid *search engine advertising,* such as sponsored products, to help sellers gain top positions in search results. Additionally, sellers can use *search engine optimization* techniques to improve their organic rankings by aligning the product listings with Amazon's ranking algorithm [33]. However, Amazon does not publicly disclose the full details of its ranking algorithm, leaving sellers to speculate about which product listing features most affect rank position. Furthermore, ranking positions are highly dynamic and influenced by evolving user behavior and competition. As a result, while achieving a top rank is one challenge, maintaining it over time presents an additional and ongoing difficulty for sellers.

Hence, this empirical research uses hourly scraped data from **Amazon.be**, comprising 28,406 product listings, to study Amazon's ranking algorithm and provide sellers with actionable insights. The data and code are publicly available in the GitHub repository [1]. A **learning-to-rank (LTR)** machine-learning model is used to approximate Amazon's ranking algorithm and analyze the importance

[1] https://github.com/thaochithai/amazon-ltr-survival-analysis.

of various features in determining product positions. Building on this, **survival analysis** is applied to examine which features affect the stability of product rankings over time. The key contributions are as follows:

1. **Empirical insights** into Amazon's product ranking algorithm are offered by replicating its ranking model and, from there, analyzing the contribution of each feature to ranking performance.
2. **Survival analysis** is introduced as a novel approach to evaluate the temporal stability of product rankings.
3. **Actionable recommendations for sellers** are proposed by identifying features that influence not only a product's initial ranking but also its ability to sustain a high position over time.

In what follows, the research process is described in detail over six sections, each highlighting a specific aspect of the approach taken:

1. **Literature Review** examines existing research on e-commerce search engines, Amazon's ranking algorithms, and survival analysis for ranking stability. It also reviews optimization strategies from both the sellers' and platforms' perspectives.
2. **Methodology** describes the research design used to address the research questions. It also covers the data collection process and explains the rationale behind the modeling choices and deployment approach.
3. **Results** present findings from the learning-to-rank models and survival analysis.
4. **Discussion** interprets the results and compares them with prior research.
5. **Critical Reflection** outlines practical implications for sellers. It also discusses limitations and suggests directions for future work.
6. **Conclusion** summarizes the research findings and implications.

1 Literature Review

This section discusses e-commerce search engines and their information retrieval architecture to lay the foundation for how search results are returned from a query. Following this, research on e-commerce search engines is reviewed, specifically focusing on Amazon's ranking algorithm and product ranking optimization techniques.

1.1 Search engines

Searching and retrieving information from information resource systems using a query is regarded as information retrieval [13]. The type of information retrieved can be text, images, and videos, and the applications used to retrieve information are typically search engines [13]. Information retrieval systems, such as search engines, not only return matched results but also rank them by relevance to queries.

The three-step information retrieval process of search engines Search engines operate through crawling, indexing, and ranking [13], illustrated in Figure 1.

1. *Crawling*: automatically discovers and downloads web pages [13]. This step enables search engines to compile a large collection of documents and web content.
2. *Indexing:* organizes and categorizes the content in a structured format, namely indexes. In this way, when users enter queries, the search engine can easily search the collected web content [13].
3. *Ranking*: orders matched results based on query relevance [13,33].

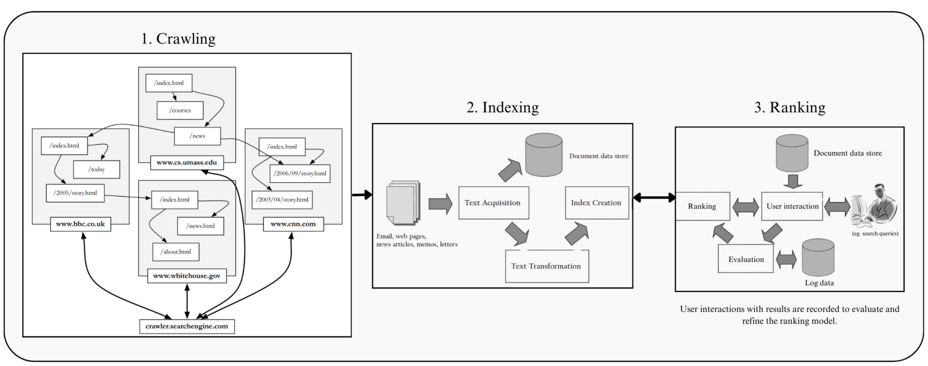

Fig. 1: Three-step process illustration is adapted from Crof et al.[13].

Types of search engine Search engines can be categorized based on the intent behind the query, such as informational, navigational, and transactional [25,48,13] (see Appendix A for examples).

- *General web search engines* index large amounts of data from various sources across the Internet. They provide broad search results that address all user intents [33].
- *Vertical search engines* focus on specific intent or industries and deliver results tailored specifically to those domains [13,33].

1.2 E-commerce Search Engines

E-commerce platforms such as Amazon, eBay, and Alibaba are vertical search engines specializing in product search [58]. For example, a search for 'air fryer' on a general search engine such as Google or Bing might return a mix of websites, product reviews, news articles, and product listings from e-commerce sites. On an e-commerce search engine, the results page would solely consist of product listings. E-commerce search engines adopt the same three-step process as general search engines but refine it to meet the specific needs of their domain. Since sellers upload their products directly onto the platform, there is no need to crawl the internet to acquire product information. E-commerce search queries on Amazon typically consist of 2-3 words that focus on the product's attributes [37]. Unlike web queries, which are usually natural language sentences, e-commerce queries thus specifically target product properties [28]. To accommodate short queries, e-commerce search engines implement specialized retrieval and ranking processes that effectively interpret user intent and return relevant products [37].

E-commerce product retrieval E-commerce product retrieval uses a multi-tier information retrieval architecture common to modern search applications [14]. Figure 2 shows this four-stage retrieval process. Each stage is explained using the following example queries: "air fryer," "fryers," "air fryer for the large family," "black fryer," and "Phillips air fryer."

1. **Query understanding** identifies the products, categories, and attributes users search for [38,37]. This is done through:
   - **Language normalization** standardizes queries by correcting misspellings, capitalizations, and plurals [14]. For example, "Philips air friers" is normalized into "philips air fryer".

  - **Intent classification** identifies the product users are searching for [38]. The user is looking for air fryers within the Home & Kitchen category in all the example queries. It also breaks down the query to identify product attributes such as color, brand, or category [37]. For example, “black air fryer” reveals a preference for a color, "Philips air fryer" for a brand, and "large family" shows a large capacity need.

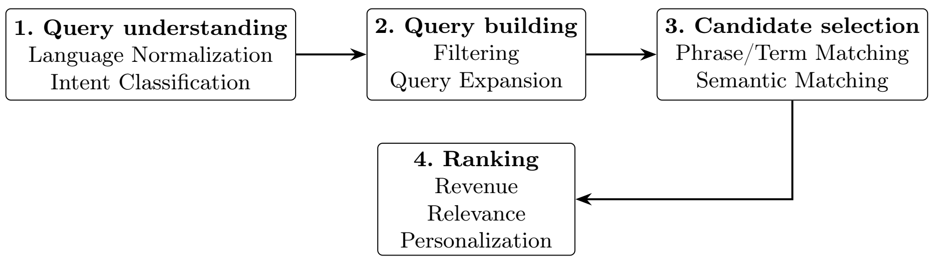

Fig. 2: Multi-tier information retrieval of a search application [14]

2. **Query building** modifies or reformulates the query to improve search results [4,24]. This is achieved by both:
   - **Filtering** removes irrelevant or non-informative words, also referred to as “stop words”, from the query [4]. For example, in the query “air fryer for large family,” “for” is removed.
   - **Expanding** includes synonyms, related terms, or spelling variations to enhance results [4]. To illustrate, "air fryer basket" is a term related to the example queries and ,thus, an expansion.

3. **Candidate selection** matches queries with potential products [14]. This is realized by:
   - **Phrase/term matching** matches the term in the query with product information [35]. This approach is limited when product listings do not contain the exact query [35].
   - **Semantic matching** uses machine learning to interpret the meaning behind a query and product listing [35]. For the example “air fryer for a large family”, the model interprets the meaning and matches it to larger capacity air fryers, even if “for a large family” is not mentioned in the product listings.

4. **Ranking** models are optimized for multiple objectives, such as **relevance**, **revenue**, and **personalization** [14], which are discussed in detail in the following section.

1.3 Amazon search and product ranking

Amazon does not publicly disclose the full details of its product ranking algorithm. However, official Amazon sources, such as papers on **Amazon Science** [67,59,14,43,22] and blog posts for Amazon sellers on `sell.amazon.com`, offer valuable insights into Amazon's ranking models and features. Additionally, existing research papers on e-commerce ranking on Google Scholar are also reviewed.

Amazon ranking objectives Amazon pursues multiple objectives [43]. Research on Amazon Science indicates that the ranking model aims to maximize:
- Relevance (R) [59,10]
- Purchase likelihood (P) [59,10]
- User engagement (E) [22]

Amazon's ranking function can be defined in terms of ranking features x^2 :

$$\text{Rank score} = f(x) = \max(\text{objective}(\mathrm{R}, \mathrm{P}, \mathrm{E}) \tag{1}$$

Amazon ranking features E-commerce platforms rank products using a score calculated by a function with various input features [49]. This function is trained to predict ranking scores by optimizing specific objectives such as relevance [49,21]. The input features used in the ranking process are typically categorized into two groups [21]: **content-based features** (e.g., product title, description, color, brand) and **behavioral features** (e.g., clicks, views, purchases). For Amazon, these input features are slightly different. Sorokina and Cantú-Paz [59] were the first to reveal the Amazon ranking features:
- **Behavioral features** are related to users' interactions with the product listing (e.g., clicks, views, add-to-cart, purchases).
- **Product features** are related to product listing attributes (e.g., sales data, reviews, and product popularity).
- **Query features** are context and semantic aspects of the query. Amazon parses query attributes (e.g., color, type, brand) and identifies the implicit intent or context (e.g., "Philips air fryer" is identified as an air fryer) to better understand searches [37].
- **Textual similarity** measures relevance between the query and the product's textual features. Specific metrics are not disclosed.

[2] When mentioning x in a formula later, it is understood as feature input.

Among these features, behavioral signals are considered the most important [59], particularly conversion rate [22]. However, for new products with limited user engagement, relying solely on behavioral features can lead to poor ranking [22]. To address this initial lack of behavioral data (cold start problem) of Amazon, Gupta et al. [22] developed a method that uses historical data from similar products, with features such as brand popularity, seller past performance, and product type, to predict engagement for new products. *Beyond Amazon Science research*, many studies also examine Amazon's ranking algorithm and identify additional features such as click-through rate (CTR), conversion rate (CR), product sales [40], reviews, average rating, and price [49]. In 2019, reports claimed that Amazon shifted to prioritize profitability as its main objective [41]. However, Amazon clarified that they prioritize features correlated with profitability, not profitability itself [41]. The feature is speculated to be contribution profit (dollar amount each product adds to overall profit) [41].

1.4 Products ranking models

Learning to Rank (LTR) models are commonly used in commercial search engines, where e-commerce marketplaces rank products based on their relevance to a query [36,29,28]. LTR learns a ranking function using features extracted from query-product pairs [21,39]. A ranking function calculates a relevance score for each product, which determines their order in the search results, leading to an ordinal rank [21]. These functions can range from simple linear models to advanced machine learning or neural networks [49]. Currently, machine learning through discriminative training based on user feedback, such as purchases, clicks, or views, is the state of the art for LTR [36]. Discriminative training relies on users' feedback, such as purchases, clicks, or views. For instance, if users often purchase product A but only view product B for the same query, the model learns to rank product A higher, reflecting its greater relevance [36]:

- **Pointwise LTR** evaluates the relevance of a query to product listings individually [36]. It functions similarly to classification or regression models that predict the absolute relevance score or classify whether a product is relevant or irrelevant [28].
- **Pairwise LTR** does not consider the individual relevance score of the query to the product [28]. It compares pairs of products

and learns which one should be ranked higher. This comparison is repeated for all the retrieved products to create a rank [28,36].
- **Listwise LTR** analyzes the entire list of retrieved products, predicting the best order in one step to match the ideal relevance objectives, which sorts from highest to lowest relevance [28,36].

Pairwise and listwise approaches are widely adopted in e-commerce ranking problems because they focus on comparing products relative to one another [9].

Gradient Boosted Trees and LambdaMART Gradient Boosted Tree (GBT) is a classic yet highly accurate and effective machine learning approach for LTR problems in online marketplaces [39,23]. GBT builds sequential decision trees, with the new tree focusing on correcting errors made by previous trees [46]. LambdaMART [7] extends gradient boosting specifically for ranking problems. Unlike regression models that predict absolute values, LambdaMART learns the relative ordering of items by considering how swapping pairs of items would affect ranking quality [28]. The algorithm optimizes using a gradient (which points the algorithm toward better predictions), computed from the normalized Discounted Cumulative Gain (nDCG) metric [8]. The DCG at rank position k measures ranking quality by rewarding placing the relevant item at the top:

$$\text{DCG@k} = \sum_{i=1}^{k} \frac{2^{l_i} - 1}{\log_2(i+1)} \tag{2}$$

where l_i is the relevance label of the document at position i. The numerator $2^{l_i} - 1$ assigns higher weight to more relevant items, while the denominator discounts lower positions[3] [8]. To normalize DCG across queries with different numbers of relevant documents, nDCG is divided by the ideal iDCG[4]:

$$\text{nDCG@k} = \frac{\text{DCG@k}}{\text{iDCG@k}} \tag{3}$$

[3] The relevance label (l_i) determines contribution, e.g., relevance label 3 yields 7 points ($2^3 - 1$), while label 1 yields 1 point ($2^1 - 1$). This contribution is then discounted more heavily as i increases (or lower position). Thus, the ranking quality is ensured as the model rewards placing higher relevant product to higher position or lower i.

[4] The ideal iDCG is the maximum DCG score obtained when products are perfectly ordered by their ground truth relevance labels l_i.

For a set of queries $Q = \{q\}$ and products $d_k \in I_q$, LambdaMART learns a scoring function f that maps an input feature vector to a relevance score $s_k^q = f(x_k^q)$. The model considers pairwise relationships $(k, j) \in P_q$, where product k is more relevant than product j for query q. The pairwise loss function is:

$$\mathrm{cost}(s_k^q, s_j^q) = |\Delta\mathrm{nDCG}_{k,j}^q| \cdot \log\left(1 + e^{-\sigma(s_k^q - s_j^q)}\right) \tag{4}$$

LambdaMART minimizes this loss using GBT trained on lambda:

$$\lambda_{kj}^q = -\sigma \cdot |\Delta\mathrm{nDCG}_{k,j}^q| \cdot \frac{1}{1 + e^{\sigma(s_k^q - s_j^q)}} \tag{5}$$

Amazon ranking models Similar to the e-commerce ranking models, Amazon uses LTR models as a core rank approach [22,59,43]. Sorokina and Cantú-Paz [59] and Gupta et al. [22] used GBT with pairwise ranking objectives and optimized using nDCG. In another paper on Amazon Science addressing the multi-objective relevance ranking, the LambdaMART algorithm is used as a baseline model to gradient lambda with nDCG as the primary cost, and for each additional objective, a new gradient lambda is computed [43] .

1.5 Ranking survival analysis

E-commerce platforms, including Amazon [22,59], regularly update their ranking models based on user behavior and feedback [36,29], making product rankings dynamic. For sellers, it is important to know which features contribute to reaching top positions and which factors help products maintain their ranking over time. Several studies examined "rank survival" to identify the factors that affect how long items stay in ranked positions, such as in music album charts or app store rankings [6,27]. However, research on rank survival for e-commerce product listings, especially on Amazon, is limited. Bhattacharjee et al. [6] analyzed the Billboard 100 album chart to see which albums survive. They incorporate a stochastic process model, Kaplan-Meier Estimator, and Weibull Proportional Hazards Model (PHM) to capture rank transitions over time [6]. Jung et al. [27] examined product survival on the Korean App Store top 100 chart and also applied the Weibull PHM. For e-commerce recommendations, Wang and Zhang [64] proposed a model that incorporates timing into the recommendation process. They introduced an opportunity model based on the PHM to estimate the likelihood of an user purchasing a product at a particular time in the future. This model is built on the idea that recommending the right product is not enough, as when the recommendation is made is also crucial. For instance, if a user just bought a laptop, it would be ineffective, and possibly annoying, to recommend another laptop immediately after the purchase [64].

Kaplan-Meier Estimator This is a non-parametric method, widely used to study the time-to-survival data[5] of subjects [16]. The cumulative probability that a subject survives up to time t [16]:

$$\hat{S}(t) = \prod_{j:t_j \leq t} \left(1 - \frac{d_j}{n_j}\right) \tag{6}$$

where:
- t_j : time points of j-th events (deaths)
- d_j : number of events (deaths) that occurs at time t_j
- n_j : number of subjects at risk (have not experienced the event yet) just before time t_j

Weibull Proportional Hazards Model The proportional hazards model (PHM), is one of the most widely used parametric methods for analyzing survival data. The model estimates a hazard function that has two components [47]:
- The *baseline hazard function* represents the risk of the event occurring at a time t when all covariates are zero [47]. Weibull PHM assumes the function follows Weibull distribution: $\lambda_0(t) = \lambda p t^{p-1}$, where p is the shape parameter that determines how the hazard changes over time, making the model parametric.
- The *covariate function* acts as a scaling factor that proportionally modifies the baseline hazard, regardless of time [47]. It determines the magnitude of change in the hazard associated with different covariate values [47]. The effect is typically modeled in a log-linear form with covariates Z_i and their coefficients β_i, as $\exp\left(\sum_i \beta_i Z_i\right)$.

[5] The survial analysis method is originally developed to study patient survival times in medical research. In this context, "survival" refers to the probability of a subject (patient) remaining in a given state between time t and just before time t. The estimator computes the survival probability at each observed event time, updating the estimate step-by-step until the occurrence of the event (death) [16].

$$\lambda(t; Z) = \psi(Z) \cdot \lambda_0(t) = \exp\left(\sum_i \beta_i Z_i\right) \cdot \lambda p t^{p-1} \quad (7)$$

where:
- $\lambda_0(t)$: baseline hazard function when all $Z_i = 0$
- $\psi(Z)$: multiplicative effect of the covariates, $\exp\left(\sum_i \beta_i Z_i\right)$
- β_i: coefficient for covariate Z_i
- Z_i: value of the i-th covariate

When exponentiated, covariate term yields a hazard ratio, which quantifies effects of a one-unit increase in Z_i on the hazard rate [47]. If $\psi(Z) > 1$, the covariates increase the hazard (i.e., the risk of event occurrence increases); if $\psi(Z) < 1$, they decrease the hazard.

1.6 Ranking Optimization

For sellers, improving product rankings and achieving higher positions on SERPs leads to greater visibility, increased user engagement, and higher sales conversions [61]. However, few studies offer sellers practical insights on how to optimize their products' SERP positions. Product rank optimization conceptually resembles optimizing webpage rankings on general search engines. Therefore, similar to webpages, sellers can use two strategies to boost product rank on e-commerce SERPs: *Search Engine Optimization*, achieving top positions organically; and *Search Engine Advertising*, paying for ad placement so that products appear in the top positions.

Search Engine Optimization (SEO) For web content, SEO includes both on-page strategies, like improving content quality, and off-page strategies, such as building hyperlinks [13]. General search engines like Google use both query-independent signals, like PageRank, which assesses webpage importance based on link structure and authority[36], and query-dependent signals that measure relevance to specific search terms [45]. In contrast, e-commerce platforms like Amazon focus primarily on query-product-dependent features, including sales history, customer behavior, and textual relevance [43]. Since products on these platforms are directly listed by sellers rather than discovered through crawling, link-based SEO factors are irrelevant. Traditional web SEO strategies, hence, do not fully apply.

On-page optimization Keywords are a key component of SEO, as product listings must include terms that match user queries in order to appear in search results [2]. Amazon's seller blog recommends using 10–20 keywords per listing, balancing short-tail (specific) and long-tail (broad) keywords to capture various search intents [2]. For effective optimization:

- **Product titles**: include key attributes like brand, model, series, size, color, and capacity [42].
- **Bullet points**: provide at least three informative bullets covering usage, key features, dimensions, age group, and capacity [42,2]
- **Product descriptions**: reinforce keywords and detailed information to enhance relevance [42,2]

Off-page optimization Amazon sellers' blog suggests sellers add backend search terms [6] to enhance discoverability [2]. Backend keywords should avoid repeating on-page keywords by incorporating abbreviations, alternate names, and latent semantic indexing (LSI)[7] [2].

Sales optimization plays a critical role in Amazon's ranking, as the algorithm aims to maximize purchase likelihood [67]. When product A receives more purchases than product B for the same query, it is deemed more relevant and ranked higher. Numerous studies have examined factors that influence sales performance on Amazon, such as *review volume* [34,61], *ratings* [61], *number of Q&A entries* [17], *price* [61], and *Amazon Prime* [18]. Especially for price, Chen et al. [11] found that sellers using algorithmic pricing[8] achieved higher conversion rates and more customer feedback, even without being the cheapest. Amazon itself encourages sellers to increase sales by earning the Prime badge and offering competitive pricing [1].

Search Engine Advertising (SEA) is particularly effective for new products, which typically lack sales history, reviews, and organic ranking. Newly launched products with sponsored labels are 231 times more likely to generate ad-attributed sales than old product

[6] Backend keywords are terms hidden from customers but used by Amazon's search engine for indexing [2].

[7] LSI is including related keywords to help search engines understand the content context and match a wider range of search queries [2].

[8] Sellers use computer algorithms to dynamically adjust the prices of their items to be more price competitive [11].

listings on Amazon [1]. For example, Amazon's sponsored products is an SEA solution that gives products immediate visibility by placing them in prominent positions on SERPs. The specific ads shown on SERP are determined by the relevance between the user's search query and the keywords targeted in the advertising campaign [3]. Sponsored ads operate on a pay-per-click (PPC) model, meaning sellers are only charged when a user clicks on their ad [3]. Sellers bid on keywords to secure a sponsored position on SERP. Therefore, *effective keyword targeting*[9] *and bidding strategies [3]* are crucial for SEA success. Effective bidding strategies should carefully balance cost and profitability. Amazon allows sellers to adjust bids to 900% to remain competitive for top-of-search on the first page [1].

Most existing research addresses ranking from the platform's perspective, focusing on developing and enhancing learning-to-rank models to optimize search result pages [28,29]. In contrast, there is limited research aimed at supporting sellers in improving the rank of their product listings. Although some studies have investigated Amazon's Best Seller Rank, or sales rank, to guide sellers on improving sales performance [18,34,56], several research gaps remain:

1. The majority of studies prioritize algorithmic or model optimization for marketplaces, rather than providing sellers with actionable SEO or listing optimization strategies.
2. While platforms like Amazon disclose some ranking factors, many critical signals remain proprietary. Moreover, sellers often lack insight into the relative importance or weight of these factors in the ranking process.
3. Empirical research on search engine optimization practices specifically tailored to Amazon is scarce; most available knowledge comes from informal sources such as Amazon Seller pages [2,1,3] and seller blogs [42,40].
4. No existing research has explored how long products persist in search rankings over time (product rank on e-commerce SERP) or which factors influence the extension or reduction of this survival time within e-commerce SERPs.

[9] Amazon advertising offers two distinct keyword targeting strategies: automatic targeting by algorithm or manual targeting by sellers

2 Methodology

2.1 Research questions and emperical design

To address the gaps identified in the literature review, this research investigates two questions to provide sellers actionable insights:
– **RQ1**: How does Amazon assign ranking to products on its SERPs, and which features contribute most significantly to ranking?
– **RQ2**: What factors help products maintain their rank on Amazon SERPs over time?

These questions guide a two-part analysis: (1) building a ranking model and identifying the influential ranking features and (2) analyzing the stability of products' ranking over time. This section describes the overall methodology, data collection, cleaning and preprocessing, model selections, implementation, and evaluation.

2.2 Data collection

Data collection involved web scraping key features based on those identified in the literature [59,49], including product title, price, reviews, rating, Prime badge, sponsored status, sales badges, Best Seller Rank (BSR), bullet points, and description, according to the set of features defined in the literature review [59,49]. The scraping process consisted of three steps illustrated in Figure 3:

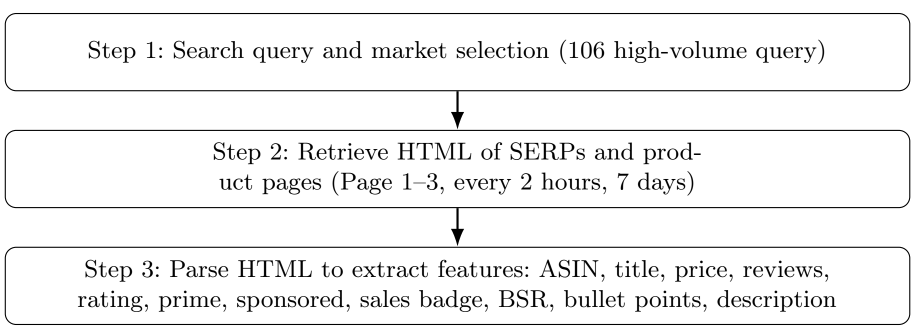

Fig. 3: Three-step data collection

Step 1: Search query and market selection Search terms are collected from the Q1 2025 agency report [19], identifying 120 popular Amazon keywords by search volume. This query list is selected due to keyword variations spanning broad categories, including both long-tail and short-tail ones (e.g., for query "toothbrush", there are "electric toothbrushes" and "Oral-B electric toothbrush"). After filtering irrelevant terms[10], a final set of 106 keywords remained (see the keywords used in Appendix B). Since Amazon's multilingual search model ensures no discrepancy when using English queries across locations [53], Amazon Belgium (Amazon.be) is chosen given the convenience of research location.

Step 2: Retrieve HTML of search result pages A scraper is developed in Python using standard web scraping libraries. Selenium [51] retrieves HTML content, while BeautifulSoup [55] handles parsing and extraction (see details of the scraping set-up in Appendix B). Scraping is scheduled every two hours for 7 days from 21 to 27 March 2025[11] to cover weekends and weekdays. The process returned 1,602,720 HTML files.

Step 3: Parse HTML to extract structured information Product listing features, ranking positions, scrape date, and time are extracted and combined in tabular format with the product ASIN as an unique identifier below.

2.3 Data cleaning and pre-processing

Initial cleaning Missing ASINs are dropped [12], leading to the final dataset of 1,524,775 rows with 28,406 unique product listings.

String variables are transformed to numerical or binary variables except for *ASIN* and *query* which are used as identifiers. BSR strings are parsed into two integer variables: *main category sales rank* and *sub-category sales rank*. Missing BSR values, indicating unranked products, are assigned a value of 9999999. Sales history is binary-coded (0,1) into sales badge, to show whether the product had sales in the past month or not. Text fields, including *title, description, and bullet points*, are preprocessed by trimming whitespace, converting to lowercase, identifying unusual patterns, and removing text entries where sellers put random characters to bypass

[10] Queries such as 'kindle unlimited free books' or 'gift card' introduce noise to model training, as search results returns are typically narrow and less relevant products, making it difficult for the model to learn generalizable ranking patterns.

[11] These dates are arbitrary, chosen within research phases without public holidays and sales seasons to avoid seasonal effects.

[12] ASINs have no HTML content due to timeout requests. As the same set of queries is repeatedly scraped, the number of unique files is smaller than HTML files.

requirements. Amazon uses both textual (e.g., phrase/term matching) and semantic matching to determine query–product relevance [14]. Following this, three similarity measures are computed for each query–product pair, by considering a product's title, bullet points, and description:

- Term frequency (TF)[13]
- Term frequency-inverse document frequency (TF-IDF)[13]
- Sentence-BERT (sBERT)[14]

Table 1: Parsed fields and definitions

Field	**Type**	**Definition**
asin	string	Amazon Standard Identification Number
query	string	The search query used to find the product
scrape date	date	The date the data was scraped
title	string	The title of the product listing
bullet points	string	Key highlights of the product as listed by the seller
description	string	Detailed description of the product
sponsored	boolean	1 = sponsored (ad placement), 0 = organic
sales history	object	Past month's sales ("50+ bought in past month")
prime	boolean	1 = Amazon Prime, 0 = non-Prime
bsr	string	Sales rank in main and sub-category ("206 in Home & Kitchen, 2 in Air Fryers")
page number	integer	SERP page number
position	integer	Position on the search results page
scrape hour	integer	Hour of the day when scraping occurred
price	float	Current product price
original price	float	Original product price
reviews count	integer	Total number of reviews
rating	float	Average customer rating

Numeric variables such as *page number, position, and scrape hour* are retained as-is. Price, reviews, and two new sales rank variables are log-transformed for normalization. Two additional variables are derived from price: *discount*, a binary indicator of whether a discount is present (if original price is available), and *price competitiveness*, calculated as the relative (inverted) price rank within the same query group. A new binary *popularity* variable is also created, based on whether listings have sales history, reviews, and sales rank

[13] TF and TF-IDF capture word-level similarity by measuring how often query terms appear in product listings and how informative those terms are across listings [52].

[14] sBERT is a pre-trained embedding model that captures sentence-level semantic similarity [54]. Model is loaded from: https://huggingface.co/sentence-transformers

within the top 20%, following definitions from prior work [29,50]. The details are available in Appendix C).

Data pre-processing for each model Two distinct datasets are created, each tailored to the specific purpose of the employed models (see detailed descriptions of model-specific variables in Appendix C).

Ranking model Sponsored product listings are excluded from the dataset as these are not algorithm-ranked and vary across search terms and time. The final dataset comprises 1,225,377 rows and includes the following variables:

- **Relevance label - model target**: Amazon likely uses purchase, revenue, or CTR as training targets; however, this behavioral data is not accessible. Therefore, products' actual rank on Amazon is converted into relevance labels (1-5, higher means more relevant) through quantile-based binning within each query group, assuming these ranks reflect true relevance to search queries.
- **Input features - model predictors** are sales rank, sponsored, review, sales badges, rating, Prime, discount, popularity, price competitiveness, and textual and semantic similarity.

Ranking survival model The dataset includes the complete data with the following variables:

- **Survival time - model target** refers to the duration a product stays within a defined rank threshold. *Survival hours* capture the longest uninterrupted period a product remains at the same or a better rank, checked every 2 hours. *Survival days* measure the same concept but at a daily granularity. A new streak begins whenever the product drops to a lower rank.
- **Input features - model predictors** include initial rank, sales rank, sponsored, review, sales badges, rating, Prime, discount, popularity, and price competitiveness.

2.4 Models Development

Overview of methodology Two models are used to answer two research questions.

- For RQ1, a ranking model is built from the input features, evaluate the impact of each feature on rank.
- For RQ2, survival analysis is used to analyze product survival in a rank threshold and how input features impact survival time.

Ranking model

Model implementation Extreme Gradient Boosting (XGBoost) [65] library in Python is implemented to train the ranking model. Specifically, XGBRanker is used with rank:ndcg objective based on the LambdaMART algorithm [65], given that Soriki et al. applied GBT with the nDCG objective for the Amazon rank model [59].

Data splitting Data is split into 80% training and 20% testing sets based on query ID. Out of 106 queries, 22 queries are randomly selected for the test set. This approach ensures that the model learns from complete sets of queries and is evaluated on unseen queries. To ensure comprehensive model evaluation, 5-fold cross-validation is implemented, randomly dividing queries into five subsets. The model is iteratively trained on four subsets while testing on the remaining subset. Cross-validation was applied on the 80% training set to tune hyperparameters and identify the best-performing model. This optimal model was then evaluated on the remaining 20% test set to assess its generalization performance.

Hyperparameter tunning Initially a basic ranking model is trained using the default set-up to optimize nDCG@k outlined in XGBoost LTR documentation [65] with `rank:ndcg` as training objectives, `topk` as pair generation method, and the number of pairs compared per sample as k [65]. While this offers an initial starting point, achieving strong generalization performance typically requires iterative hyperparameter tuning [65]. Random search is used to find the optimal parameters (see details in Appendix D).

Model evaluation **nDCG@k** is used to assess how well the model predicts the assigned relevance labels. As nDCG rewards placing highly relevant items near the top positions, a higher score indicates better ranking quality [8]. However, nDCG only specifies how well the model places more relevant products but does not indicate how many of them are also in Amazon's top-k listings. Therefore, **Precision@k**, adapted from Mean Average Precision (MAP) [29], is used to measure the overlap between predicted and actual top rankings. The precision@k is calculated as the proportion of items that appear in both the model's top-k predictions and Amazon's top-k ranked items. For instance, if a model predicts items [A, B, D, C] as its top-4 while Amazon ranks items as [A, B, C, E], then the precision@4 would be 0.75 since only items A, B, C appear in both top-4 lists. This enables evaluating the model's ability to predict the top k products and how closely the predicted ordering matches Amazon's actual rank.

Feature importance evaluation XGBoost's built-in importance scores (XGBoost Gain) and mean absolute SHapley Additive exPlanations (SHAP) values are used to evaluate the overall impact of each feature on model performance. Higher weights mean that the feature is more predictive of the relevance score (ranking). Moreover, SHAP values of individual predictions are analyzed to determine whether features have positive or negative effects on specific predictions.

Ranking survival model

Models selection Kaplan-Meier survival curves are applied to estimate survival probability. Weibull PHM is used to determine which product features influence survival time. Weibull is chosen over Cox PHM, as several papers have demonstrated its effectiveness for studying rank survival [6,27]. Furthermore, products might not experience the event during the observation period. This is referred to as censored data, occurring when products either still remain in their ranks at the end of study period (right-censored) or enter the dataset after the observation has begun (left-censored) [32]. Weibull can efficiently handle censored data while providing reliable β_i estimates [27].

Implementation design **Survival** is defined as the duration a product remains at or above the initial rank. The **event** of interest is when it drops below that rank (product "death"). To suggest insights for both SEO and SEA strategies, organic and sponsored listings are analyzed separately to compare potential differences.

Survival time evaluation Survival probabilities are analyzed by observing the Kaplan-Meier survival curves over survival time. Higher curves indicate better survival probabilities, while steeper drops reveal periods of rapid threshold failure. For the Weibull PHM, model coefficients are exponentiated to obtain hazard ratios for interpretation of associated risks.

3 Results

This section presents the ranking model and survival analysis results. Ranking model performance provides the basis for assessing the reliability of the subsequent feature importance interpretation. A model that closely approximates Amazon's actual ranking behavior suggests that identified features genuinely matter. Subsequently, the survival analysis explores the temporal stability of product rankings. It reveals which factors contribute to greater ranking persistence and which are associated with faster drop-offs.

3.1 Results of LTR Model

Table 2 presents the five-fold cross-validation results of the ranking model, evaluated before and after hyperparameter tuning. Based on randomized search, the optimal configuration selected was the `mean` pair generation method with 16 LambdaRank pairs per sample[15]. This strategy generates more effective pairs for training compared to the original `topk` method, providing better generalization across different ranking positions [65].

Table 2: 5 cross-validation results of model before and after tuning

@k	**nDCG**		**Precision**	
	Before tuning	Tuned	Before tuning	Tuned
5	0.880	0.899	0.351	0.347
10	0.859	0.874	0.414	0.419
20	0.827	0.840	0.465	0.475
30	0.807	0.823	0.502	0.525
Avg	0.843	0.859	0.433	0.441

Notably, nDCG@5 increased from 0.880 to 0.899, and the overall average nDCG rose from 0.843 to 0.859. This shows that, on average, the model correctly places 85.9% of items corresponding to Amazon as the ground-truth relevance. In other words, it effectively places highly relevant items (label 5) near the top and pushes less relevant ones (label 1) to the bottom. While the overall average

[15] Other tuned parameters include a learning rate of 0.186, a maximum tree depth of 5, 161 estimators, a subsample rate of 0.551, a column sampling rate of 0.832, and a minimum child weight of 2.

precision was modest (0.441), the precision increased substantially from 0.347 at k=5 to 0.525 at k=30. This indicates that, in the top 30 positions, 52.5% of the model's predicted product ranks exactly match Amazon's actual ranking. This suggests that the model captures broader relevance patterns effectively, even if precise ordering of highly ranked items presents a greater challenge. *Overall, this result highlights the model's effectiveness in approximating Amazon product ranking.*

Per-query Performance Table 3 presents the model's performance across different queries. The top and bottom performers are determined based on their *average nDCG and precision* scores. **Top-performing queries** achieved high average nDCG scores above 0.945. Precision measures the percentage of correctly ranked items among the top k, while the average actual rank@k indicates how far the predicted ranks on average, deviate from the ideal positions.[16]

For the top three queries, the average precision indicates that at

Table 3: Per-query model performance

Query	Avg. nDCG	Avg. Precision	Avg. Actual rank@k			
			5	10	20	30
Top 5 queries						
vitamin d3	0.954	0.673	5.35	14.86	20.50	24.96
creatine	0.970	0.664	7.91	10.37	16.74	23.22
blood pressure monitor	0.945	0.648	5.14	14.87	27.49	29.12
yoga mat	0.949	0.632	7.32	11.07	25.90	36.64
bluetooth headphones	0.964	0.631	5.97	11.66	24.31	25.02
Bottom 5 queries						
iphone	0.582	0.074	69.69	72.27	69.23	70.30
vitamin c	0.600	0.132	63.85	71.05	62.90	58.44
gaming pc	0.655	0.288	49.97	55.57	65.94	68.93
tv	0.666	0.365	61.42	51.52	51.38	51.86
portable charger	0.685	0.383	59.55	53.40	47.12	48.48

least 64.8% of the top-ranked items are correctly positioned. Among the remaining items that are not exactly matched, the ranking errors

[16] For example, suppose the model predicts the top $k = 5$ products, and their actual Amazon ranks are [1, 2, 3, 10, 11]. This results in an average rank of 5.4, with 3 out of 5 items (60%) exactly matching their true positions.

are typically 2 to 4 positions from the ideal average rank at $k = 5$, 5 to 10 positions off at $k = 10$, and 10 to 15 positions off at $k = 20$ and $k = 30$. In contrast [17], **the bottom five queries** performed poorly, with average nDCG scores ranging from 0.582 to 0.685 with precision values below the overall model average and substantial deviation from ideal rank. *Overall, the results indicate that the model performs significantly better for certain search terms than others.*

Feature importance analysis Table 4 presents the feature importance results obtained using both XGBoost Gain and SHAP value approaches to assess overall feature contribution[18]. Both methods

Table 4: Feature importance using XGB Gain and SHAP values

Feature	**XGB Gain**		**Mean abs. SHAP**	
	Value	Importance	Value	Importance
Sales signals				
Sub-category sales rank	0.355	High	0.325	High
Main category sales rank	0.169	Medium	0.457	High
Sales badges	0.044	Low	0.081	Medium
Product features				
Log review	0.061	Medium	0.186	Medium
Rating	0.042	Low	0.112	Medium
Discount	0.034	Low	0.081	Medium
Log price	0.023	Low	0.195	Medium
Price competitiveness	0.030	Low	0.347	High
Prime	0.007	Low	0.001	Low
Popularity	0.004	Low	0.002	Low
Query-listing similarity				
Semantic title	0.066	Medium	0.339	High
Semantic bullet	0.027	Low	0.090	Medium
Semantic description	0.018	Low	0.079	Medium
TF bullet	0.017	Low	0.028	Low
TF title	0.017	Low	0.026	Low
TF description	0.013	Low	0.019	Low
TF IDF bullet	0.025	Low	0.037	Low
TF IDF title	0.015	Low	0.029	Low
TF IDF description	0.032	Low	0.012	Low

identify sales signals as the most influential features in determin-

[17] The average ideal rank of k=15,10,20,30 is 3, 5.5, 10.5, 15.5 respectively.

[18] The level of importance is categorized based on the value: low if importance is < 5%, medium if between 5% and 10%, and high if above 10%.

ing product ranking. Sub-category sales rank, main-category sales rank, log-transformed review count, and semantic similarity of the title emerge as key contributors in both methods. However, features such as Prime and popularity show, surprisingly, little to no impact on the model's predictions. SHAP values reveal a broader set of medium-to-high importance features, including sales badges, rating, discount, log price, price competitiveness, and semantic similarity of bullet points and descriptions. *Overall, both approaches highlight complementary aspects of feature relevance. XGBoost Gain emphasizes features that drive the largest information gain during tree splits, focusing on sales rank, reviews, and semantic title. Whereas, SHAP captures a more comprehensive view of how multiple features contribute to individual relevance predictions across the entire model.*

3.2 Results of survival analysis

The longest survival time is 164 hours when measured every two hours and up to 7 days when assessed daily[19] (see Table 8 in Appendix C). Notably, sponsored listings have higher mean survival hours with smaller standard deviations compared to organic listings on the first 2 SERPs, while this pattern reverses on the third SERP. *To recommend effective SEO and SEA strategies for sellers, we analyzed survival differences between sponsored and organic listings.*

Kaplan-Meier survival curves The log-rank test (see Table 10 in Appendix E) confirms that the survival probabilities between organic and sponsored products are statistically different at the 1% significance level (p-values < 0.01). Figure 4 illustrates the survival probabilities of products across all SERPs and for each SERP.

On all SERP pages, sponsored products start at lower survival probabilities but over time have higher survival probabilities than organic products, both when accessed on an hourly and daily basis. While organic products rapidly decline below 25% survival within the first 15 hours, sponsored products remain above 50% during this period,

[19] Survival time is defined as the longest continuous streak during which a product maintains the same or a better rank. As a result, survival time measured in days may be longer than in hours, since products might temporarily drop out at certain hours but reappear within the same day.

only dropping below 25% after hour 40. By Day 2, both types show approximately 75% survival probability. Critical vulnerability periods occur within the first 15-30 hours and between Days 2-3, with organic listings experiencing steeper declines (to 13%) than sponsored items (to 50%). Interestingly, products that survive beyond Day 3 show improved survival probability from Day 4 onward (see Fig 7 in Appendix E).

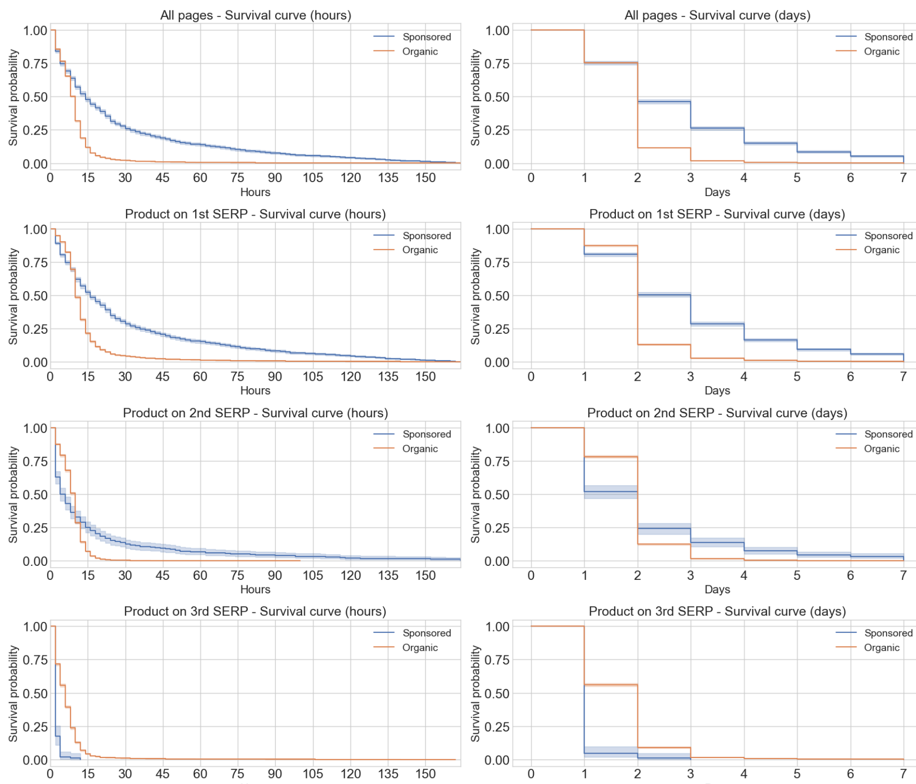

Fig. 4: Hourly and daily survival curves.

Per-SERP-page analysis Products on the 1st SERP exhibit the highest survival probabilities for both sponsored and organic listings. Sponsored products generally survive longer than organic ones on the 1st and 2nd pages. However, on the 3rd SERP, sponsored products drop sharply. None stay visible beyond 12 consecutive hours, and even if they reappear, they do not last for more than 3 days.

Weibull PHM results Weibull model coefficients are exponentiated to get the hazard ratio in Table 5 for easier interpretation of effects. A hazard ratio >1 means increased hazard rate or risk, while a hazard ratio <1 means decreased risk [62].

Table 5: Hazard ratios and Δ Risk in organic and sponsored models

Parameters	**Organic Model**				**Sponsored Model**			
	Hourly		**Daily**		**Hourly**		**Daily**	
	HR	% Δ	HR	% Δ	HR	% Δ	HR	% Δ
Prime	**0.945****	**–5.5**	**0.962****	**–3.8**	1.037	+3.7	**0.870****	**–13.0**
Popularity	**0.697****	**–30.3**	1.032	+3.2	0.815	–18.5	1.180	+18.0
Sales badge	**0.669****	**–33.1**	**0.959***	**–4.1**	0.894	–10.6	0.891	–10.9
Initial rank	**1.003****	**+0.3**	**1.002****	**+0.2**	**1.005****	**+0.5**	**1.001****	**+0.1**
Main rank	**1.059****	**+5.9**	1.001	+0.1	1.024	+2.4	**1.039****	**+3.9**
Sub rank	**0.975****	**–2.5**	1.001	+0.1	**0.966****	**–3.4**	**0.979***	**–2.1**
Price	**0.980****	**–2.0**	**0.986****	**–1.4**	0.972	–2.8	**0.958***	**–4.2**
Reviews	**0.991****	**–0.9**	1.001	+0.1	1.017	+1.7	1.008	+0.8
Rating	**1.027****	**+2.7**	0.996	–0.4	0.969	–3.1	0.977	–2.3
Discount	**0.951****	**–4.9**	1.005	+0.5	1.048	+4.8	0.941	–5.9
PC	0.964	–3.6	0.983	–1.7	**0.795***	**–20.5**	1.011	+1.1

Note: HR = Hazard Ratio; % Δ = % increase or decrease in hazard (risk of rank drop) relative to the baseline (HR = 1), *,** at the 5%, 1% levels, respectively. Binary variables: Prime, Popularity, Sales badge, and Discount (all True). Main/Sub rank = BSR in main/sub category. PC = price competitiveness.

For organic models, popularity and sales badge are the key factors, reducing the hazard of rank drop by 30.3% and 33.1%, respectively. Prime status and discount yield moderate risk reductions (5.5% and 4.9%). In contrast, a higher main category sales rank increases the hazard by 5.9%, similarly for initial rank but with a minimal effect (+0.3%). *For daily survival*, only four variables are significant, with Prime (–3.8%), sales badge (–4.1%), reducing the risk. **For sponsored models**, price competitiveness is the most influential factor for hourly survival (-20.5%). *For daily survival*, Prime status is the most impactful (–13.0%), followed by price (–4.2%) and sub-category rank (–2.1%) in contrast with main sales rank (+3.9%).

Overall, organic and sponsored products exhibit distinct survival predictors. Organic models are more sensitive to hourly changes, while sponsored models show stronger daily effects. Prime status and initial rank consistently influence all models. Notably, price and sub-rank demonstrate opposite effects between listing types.

4 Discussion

This study identifies key product features affecting Amazon rankings and ranking stability, thereby providing actionable insights for sellers to enhance product visibility.

4.1 Ranking model results

The ranking models are discussed in two aspects: (1) assessment of the model performance, as a reliable model is crucial for meaningful feature importance analysis, (2) identifying the key features and applicable seller practices.

Model performance The model in this research achieves an average nDCG of 0.859 and precision of 0.441. This demonstrates sound performance within the research field [26,29,31], especially given the reliance on publicly scraped data rather than proprietary datasets. Moreover, the performances are better for certain queries than others, which is also seen in previous research [66,31].

Feature importance While both XGBoost and SHAP offer insights, the focus is on SHAP as it reveals directional effects at the individual prediction level. Figure 8 (See Appendix E)shows a summary of SHAP value distributions, indicating the magnitude and direction of their impact.

Sales performance is the most influential in predicting relevance, with sales ranks being most important. Lower ranks increase predicted relevance, while higher ranks reduce it. This aligns with prior studies highlighting the importance of sales data in ranking [59,40]. Additionally, sales badges, which reflect monthly absolute sales (+50 to +1000 products sold), also boost relevance, but to a lesser degree. This shows the model values a product's sales within its category more than raw monthly sales, suggesting it emphasizes relative category performance over absolute sales.

Sellers should focus on boosting the sales rank more than just the absolute sales values. They should enhance factors that influence it, since rank itself is not directly controllable.

Sentimental features such as review count and product ratings moderately influence relevance predictions, thereby aligning with prior research that emphasized the role of reviews and ratings in e-commerce ranking systems [59,49]. Higher review counts and better ratings boost relevance scores. These metrics reflect customer opinions and serve as proof of purchase, thereby aligning with Amazon's objective of optimizing for purchase likelihood. In addition, these metrics also function as popularity indicators [29], as products with more reviews and high ratings typically indicate higher purchase volumes and greater popularity. However, the derived popularity feature had minimal impact, as its components (sales rank, review, rating, and sales badge) already exert strong, individual influence.

Sellers should build reviews and ratings using tools like Amazon Vine program, where trusted reviewers receive free products in exchange for reviews [20]*, or through Amazon's "Request a Review" button when customers do not leave reviews*[21]*. Managing customer concerns and promptly resolving issues also helps maintain ratings.*

Pricing Price-related features are key predictors of relevance, with SHAP analysis showing that both high and low prices can boost relevance scores depending on the context. This likely stems from price variations across search queries[22], making absolute price less reliable for capturing pricing effects. Price competitiveness (SHAP: 0.347) is more predictive than absolute price (SHAP: 0.195), as it reflects a product's price relative to others in the same query. Highly competitive prices increase relevance, while discounts have less impact. This supports findings that Amazon shoppers are less price-sensitive [12]. As a result, dynamic and competitive pricing is likely more effective than just setting the lowest price and excessive discounting.

Sellers should price competitively within queries using dynamic pricing to boost chances of getting Amazon Featured Offer [11].

Query-product similarity Lexical similarity has minimal impact on relevance predictions. In contrast, semantic similarity is highly influential, particularly in titles. While early e-commerce ranking relied

[20] https://sell.amazon.com/tools/vine
[21] https://sellercentral.amazon.com/help/hub/reference/external/G698QJEQJZFEXSAT
[22] Queries in the Electronics category have higher average basket size than in Fashion

on lexical representations, modern BERT-based models better capture semantic relationships [44]. Amazon also trained and optimized BERT-based models for ranking and retrieval [44,15]. Therefore, including and repeating keywords in titles, bullet points, or descriptions is not enough for ranking.

Sellers should focus on meaningful, semantically rich content over keyword stuffing. Amazon advises using latent semantic content [2], and tools like the Search Query Performance report to align product content with customer queries [23].

4.2 Survival analysis result

Product visibility over time is analyzed from two aspects: (1) survival probabilities of organic vs. sponsored listings, and (2) how product features affect short-term (hourly) and long-term (daily) survival.

Survival probability Distinct survival patterns emerge across listing types. *Organic products* have a sharp, early drop in hourly survival due to intense competition and reliance on strong sales signals (e.g., sales badges, sales rank). However, those that survive this early period tend to stabilize, with improved odds of survival for another hour or day (Appendix E), Figure 7). *Sponsored products*, by contrast, start with high survival probabilities due to guaranteed ad placement. Yet this advantage often fades, as many sponsored listings show flattening or gradual decline in survival probability for another day or hour (Appendix E), Figure 7). Especially those starting on SERP 2-3 are prone to early exit. This is likely due to ad pacing strategies, where sellers limit ad delivery during low-traffic or low-conversion hours to optimize ad spending[24]. Despite the difference in survival hours, both listing types converge to a 75% daily survival probability by day two. This is possibly due to listings temporarily disappearing and later reappearing.

Product features effects on survival time Feature effects vary by time frame and listing type. For hourly survival, organic products benefit most from sales badges, high ratings and review volume. These signals likely attract immediate attention. Price com-

[23] https://sellercentral.amazon.com/help/hub/reference/external/G8J4CB5ZBF3NX7TP
[24] `https://www.ppcentourage.com/posts/amazon-ad-dayparting`

petitiveness improves organic relevance but has little effect on short-term survival. In contrast, sponsored listings are more price-sensitive, with pricing reducing hourly risk. Fewer features affect sponsored products' hourly survival, likely due to ad scheduling and the cost of maintaining competitiveness across time slots. Across both time frames, initial rank and main category sales rank strongly predict survival, with lower-ranked products dropping faster. This is consistent with prior findings [6]. Interestingly, worse subcategory ranks correspond to lower hazard, possibly due to lower competition in subcategories. Price effects are complex, due to price variations across categories. For example, electronics categories typically have higher average prices than fashion.

In conclusion, the first 15–30 hours and days 2–3 are key risk periods. Surviving these windows improves ranking stability. Sellers should tailor SEO and SEA strategies to both feature-driven and time-sensitive survival dynamics.

4.3 Implications of the findings

The findings suggest that e-commerce sellers should tailor their strategies based on listing type and desired visibility timeframe. The implications are summarized in Table 6.

Table 6: Features optimization for ranking and ranking stability

Feature	**Rank**	**Hourly Survival**		**Daily Survival**	
	Importance	Organic	Sponsored	Organic	Sponsored
Sub-category sales rank	+++	↑	↑	○	↑
Main category sales rank	+++	↓	○	○	↓
Sales badge	++	↑	○	↑	○
Price competitiveness	+++	○	↑	○	○
Semantic title	+++				
Review	++	↑	○	○	○
Rating	++	↓	○	○	○
Semantic bullet	++				
Price	++	↑	○	↑	↑
Discount	++	↑	○	○	○
Semantic description	++				
Prime	+	↑	○	↑	↑
Popularity	+	↑	○	○	○
Initial rank		↓	↓	↓	↓

Note: Ranking importance: +++ = High, ++ = Medium, + = Low. Survival Impact: ↑ = Improves survival (reduces rank drop risk), ↓ = Reduces survival (increases rank drop risk), ○ = Not significant.

5 Critical Reflection

5.1 Limitations

This study has several limitations that should be considered when interpreting its findings.

- **Dataset**: this research uses public data obtained through web scraping. There is no access to behavioral data such as clicks, purchases, or other user engagement metrics.
- **Algorithmic updates**: Amazon's algorithm evolves over time, with shifts in features' weight due to updates or changing priorities. Thus, this study's findings may not fully reflect future ranking behavior. However, since the analysis targets high- and medium-impact features tied to Amazon's main objectives of relevance, user engagement, and purchase likelihood, the results are expected to stay relevant despite potential changes.
- **Generalizability**: generalizing the results to other e-commerce platforms is challenging, as each platform's algorithm has unique priorities and feature weightings. For instance, features like Prime or BSR may not apply for other platforms.

5.2 Future research

To overcome the limitations of this study and deepen insights into e-commerce product ranking, future research could benefit from:

- **Behavioral data:** incorporating behavioral data through partnerships with sellers or third-party analytics tools.
- **Categories and market segments**: broadening the scope to include more product categories would offer deeper insights into how ranking factors differ across market segments. For instance, pricing features may carry more weight in price-sensitive categories, while semantic content (e.g., bullet points, descriptions) could be more influential in feature-rich segments like electronics, where detailed specifications guide purchasing decisions.
- **E-commerce platforms**: research should extend to other e-commerce platforms, such as Walmart or Alibaba, as this would enable a comparative analysis of ranking algorithms and their unique weighting of features.

Conclusion

Online marketplaces have become essential for buyers to quickly discover products and for businesses to connect with potential customers. Ranking algorithms play a key role by placing relevant products at the top, directly affecting visibility, clicks, and sales. This makes high rankings crucial for sellers. However, existing research primarily focused on general search engine algorithms or e-commerce ranking from the platform's point of view, often overlooking the perspective and challenges faced by sellers.

This research adds a seller-focused perspective to the existing literature by analyzing key ranking factors and the survival of product listings. Subsequently, it offers insights into what drives ranking stability and success.

The feature analysis revealed that sales rank is the most influential factor in Amazon's ranking algorithm, followed by price competitiveness and semantic title relevance. Notably, semantic features were found to outperform traditional keyword-matching methods, underscoring the growing importance of contextual understanding in relevance assessments.

The survival analysis added another dimension by exploring how product rankings evolve over time. Sponsored listings exhibited higher short-term (hourly) stability compared to organic results, but they also faced a higher risk of dropping off the first three pages in the long run. These findings suggest that while paid advertising can offer immediate visibility, long-term success on Amazon requires strong organic performance, driven by product quality, competitive pricing, and customer reviews.

Looking ahead, future research could enhance the understanding of e-commerce ranking algorithms by incorporating behavioral data. Expanding the study to cover a broader range of product categories would also help determine how ranking factors vary across different market segments. Furthermore, insight into the ranking algorithms of other e-commerce platforms could lead to more generalized insights into effective ranking strategies.

Appendix A. Search engine types

General web search engines provide broad search results that address all user intents. For example, when searching for "Air fryer for large families" on Google, a general search engine, results in product listings as well as a number of sites being returned. This is illustrated in Figure 5.

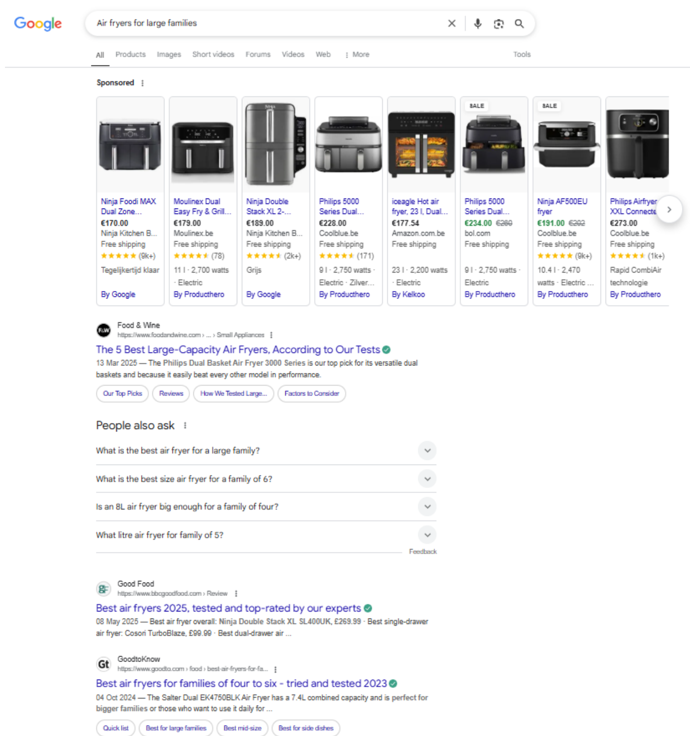

Fig. 5: General search engine: Google general results page

Vertical search engines deliver results tailored specifically to industries or intents. For example, Google Image is a vertical search engine as it only delivers results in the form of images. This is illustrated in Figure 6. for the example query "Air fryer for large families".

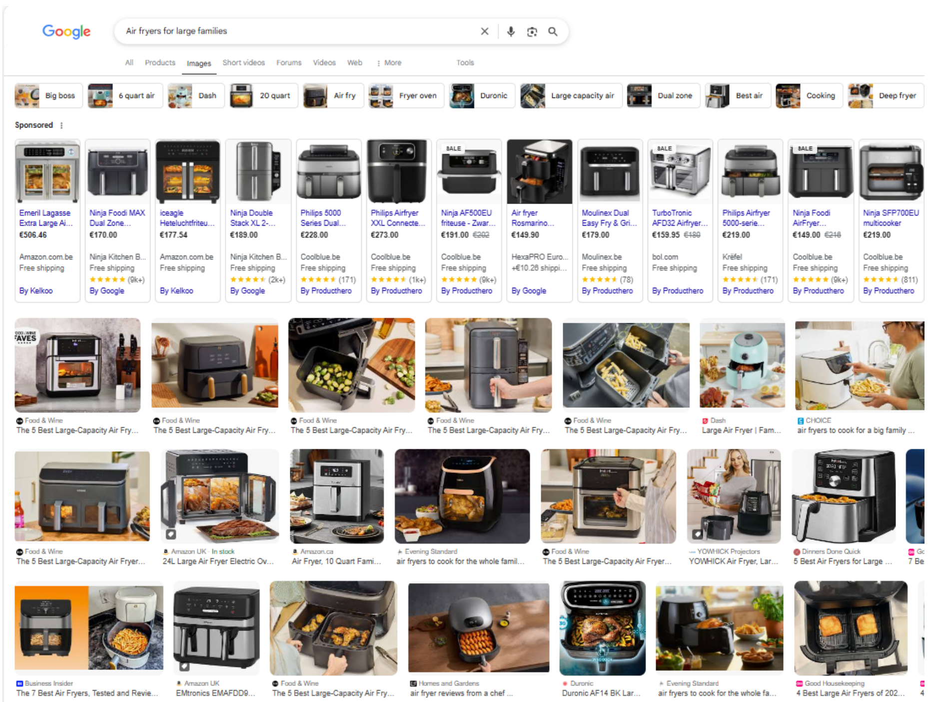

Fig. 6: Vertical search engine: Google image results page

Appendix B. Overview of scraping set-up

Queries The Table show 106 queries used to scrape the data by categories.

URL set-up To collect information from multiple pages, particularly from `Amazon.be` in English, the URL is constructed as follows:

1. For SERP HTML retrieval using search queries:

```
url = f"https://www.amazon.com.be/s?k={search_term.replace('␣','␣'+')}
&page={page_number}&language=en_GB"
```

2. For product pages HTML retrieval using ASINs:

```
url = f"https://www.amazon.com.be/dp/{asin}?language=en_GB"
```

Browser set-up Chrome is launched via Selenium WebDriver[25], with randomized user agents to mimic real user behavior. User agents are strings included in HTTP headers that identify the client's browser and operating system [20]. These were rotated to avoid triggering bot detection. Chrome is launched in incognito mode to prevent personalization. To further evade detection, we employed:

1. Chrome DevTools Protocol [26] commands to override automation-related signatures:

```
self.driver.execute_cdp_cmd('Network.setUserAgentOverride',
{"userAgent": random.choice(user_agents)})
```

2. Custom JavaScript to mask the presence of Selenium WebDriver.

```
self.driver.execute_cdp_cmd('Network.setUserAgentOverride', {
    "userAgent": random.choice(user_agents)
        })
```

3. Stochastic delays, with random pauses of 3–7 seconds between each request and a 10% probability of longer delays. Requests are executed in randomized batches.

Alternative methods that do not require launching Chrome are also tested to optimize the process. However, they are unsuccessful due to Amazon's anti-bot defenses or incomplete HTML responses, such as missing sponsored listings or blocked requests.

[25] https://www.selenium.dev/documentation/webdriver/

[26] https://chromedevtools.github.io/devtools-protocol/

Retrieval format The retrieved HTML content is saved locally to provide greater flexibility in parsing structured information and to support additional extraction needs in later stages of the analysis.

Appendix C. Dataset description & set-up

Table 7 presents a summary of all variables used in the analysis, including their descriptions, types, roles, descriptive statistics, and the models in which they are applied.

Supporting variables The initial cleaning covers the key product features that are used in both the ranking model and survival analysis. However, as each model requires a different dataset structure, additional supporting variables are needed for a model-specific set-up. These include identifiers, grouping variables, and evaluation metrics tailored to each model's requirements. The supporting variables used for dataset construction and analysis are described as follows:

Ranking model
- **Group ID** contains a unique identifier that distinguishes product listings scraped at different times for the same search query. This is achieved by transforming the search term, scrape date, and scrape hour variables into one.
- **Query ID** is a numeric value that specifies the search query used to retrieve product listings.
- **Actual rank** reflects the unique, sequential number assigned to a product listing, representing its rank among all listings for a specific search query scraped at the same date and hour. This number is assigned by sorting the dataset on query, page number, scrape date, and then scrape hour.

Ranking survival model
- **Continuous rank** refers to the rank of product listings, ranging from 1 to 180 by merging the position from three SERP pages.
- **Product-query ID** is a unique identifier that specifies a product-query pair. The variable is created by combining the search query and ASIN into one ID.
- **Initial rank** is the first observed position for each product-query.

Table 7: Description of variables used in the models

Variable	Description	Form	Descriptive statistics				Type	Role	Model	
			Mode	% frq	[Min:Max]	Mean			Rank	Survival
asin	Amazon Standard Identification	string	-	-	-	-	Original	ID	√	√
group_id	Unique identifier for product group	integer	-	-	-	-	New	ID	√	×
query_id	Unique identifier for query	integer	-	-	-	-	New	ID	√	×
product_query_id	Combined product and query identifier	string	-	-	-	-	New	ID	×	√
relevance_label	Relevance score based on actual rank	integer	-	-	-	-	New	target	√	×
survival_hours	Number of hours product survives	integer	-	-	[2:164]	71.5	New	target	×	√
survival_days	Number of days product survives rank	integer	-	-	[1:7]	1.95	New	target	×	√
prime	Amazon Prime	binary	TRUE	72.02	-	-	Original	predictor	√	√
sponsored	Sponsored product status	binary	FALSE	80.84	-	-	Original	predictor	√	√
sales_badge	Presence of sales in past month	binary	FALSE	93.50	-	-	New	predictor	√	√
discount	Product has discount	binary	FALSE	76.59	-	-	New	predictor	√	√
popularity	Product has sales badge, top reviews, sales rank	binary	FALSE	97.15	-	-	New	predictor	√	√
main_cat_rank	Sales rank in main category	integer	-	-	[1:9999999]	1410717	New	predictor	√	×
sub_cat_rank	Sales rank in sub-category	integer	-	-	[1:9999999]	1377605	New	predictor	√	×
log_sub_rank	Log-transformed sub-category rank	float	-	-	[0.69:16.12]	5.95	New	predictor	×	√
log_main_rank	Log-transformed main rank	float	-	-	[0.69:16.12]	10.27	New	predictor	×	√
log_price	Log-transformed price	float	-	-	[0.01:6.91]	3.45	New	predictor	√	√
log_review	Log-transformed review	float	-	-	[0:15.91]	5.45	New	predictor	√	√
price_competitiveness	Price competitive score	float	-	-	[0:1]	0.76	New	predictor	√	√
rating	Product rating (0-5)	float	-	-	[0:5]	4.04	Original	predictor	√	√
semantic_title	Semantic relevance of title	float	-	-	[-0.06:1]	0.5	New	predictor	√	×
semantic_bullet	Semantic relevance of bullet points	float	-	-	[-0.12:0.97]	0.38	New	predictor	√	×
semantic_description	Semantic relevance of description	float	-	-	[-0.07:1]	0.39	New	predictor	√	×
tf_title	Term frequency in title	float	-	-	[0:1]	0.14	New	predictor	√	×
tf_idf_title	TF-IDF score of title	float	-	-	[0:1]	0.28	New	predictor	√	×
tf_bullet	Term frequency in bullet points	float	-	-	[0:1]	0.05	New	predictor	√	×
tf_idf_bullet	TF-IDF score of bullet points	float	-	-	[0:1]	0.19	New	predictor	√	×
tf_des	Term frequency in description	float	-	-	[0:1]	0.06	New	predictor	√	×
tf_idf_des	TF-IDF score of description	float	-	-	[0:1]	0.18	New	predictor	√	×
actual_rank	Acutal product rank on Amazon	integer	-	-	[1:180]	-	New	evaluation	√	×
continuous_rank	Rank flaten accorss SERP pages	integer	-	-	[1:180]	-	New	-	×	√
initial_rank	Starting rank of product	integer	-	-	[1:180]	-	New	predictor	×	√

Descriptive statistics

Overall, the dataset shows that 72.02% of the product listings are eligible for Amazon Prime, and 80.84% are classified as organic. However, fewer than 7% of the listings report at least 50 units sold in the past month. Discounts are applied to 76.59% of the products, while only about 3% satisfy the criteria for popularity. On average, products have a rating of 4.0. In terms of textual relevance, search queries are more frequently matched with bullet points and descriptions, yielding higher relevance scores, whereas semantic similarity is highest in product titles. Additionally, the average rank in the main category is higher (worse) than that in the subcategory.

Regarding product temporal stability, the average survival time is 71.5 hours when measured at an hourly level and 1.95 days when measured at a daily level. Table 8 shows detailed descriptive statistics of product breakdown by listing type and SERP pages.

Table 8: Survival descriptive summary

Page	**Type**	**Survival hours**		**Survival days**		**Count**
		Mean	[Min, Max] (Std)	Mean	[Min, Max] (Std)	
1	Sponsored	29.8	[2, 164] (35.1)	2.9	[1, 7] (1.7)	3,204
1	Organic	13.3	[2, 164] (13.5)	2.1	[1, 7] (0.7)	10,287
2	Sponsored	15.9	[2, 164] (28.2)	2.0	[1, 7] (1.5)	404
2	Organic	8.9	[2, 100] (4.7)	1.9	[1, 7] (0.6)	9,158
3	Sponsored	2.5	[2, 12] (1.3)	1.1	[1, 3] (0.3)	109
3	Organic	6.8	[2, 162] (7.3)	1.7	[1, 7] (0.7)	7,580

Appendix D. Overview of hyperparameter tuning and random search set-up

Table 9 shows the hyperparameters considered for the hyperparameter tuning. Hyperparameter tuning was conducted via random search by specifying parameter distributions and randomly sampling parameter combinations rather than evaluating specific combinations as in Grid search. This approach offers better search space coverage and faster convergence compared to Grid search [5]. Key parameters that are tuned are the pair generation method and number of pairs compared [65]. Furthermore, other parameters, including learning rate, tree depth, number of estimators, subsample ratio, column sampling by tree, and minimum child weight, are also tuned in the paper applying LambdaMART for e-commerce ranking [29].

Table 9: Random search hyperparameters

Hyperparameter	**Range**	**Final selected value**
Learning rate	$\mathcal{U}(0.01, 0.30)$	0.186
Max depth	$\{3, 4, \ldots, 14\}$	5
Number of estimators	$\{50, \ldots, 300\}$	161
Subsample	$\mathcal{U}(0.5, 1.0)$	0.551
Colsample by tree	$\mathcal{U}(0.5, 1.0)$	0.832
Min child weight	$\{1, \ldots, 9\}$	2
LambdaMART-specific		
Number of pairs per sample	{8, 16, 32}	16
Pair method	{'topk', 'mean'}	'mean'

Below is a brief description for each parameter tuned:

– **Pair method**: determines the strategy the model uses to select document pairs for training. The XGBoost library offers two options: "mean" and "topk" strategies. While the "mean" strategy considers all possible document pairs within a query group, the "topk" strategy only considers the top-k products [65]. It is recommended to use the "mean" strategy for smaller datasets, as it offers more granularity and generalizability through random sampling [65]. Our tuning confirmed this recommendation, with "mean" being selected as the optimal strategy.
– **Number of pairs per sample**: determines how many pairs are generated for each document in a query group [65]. More pairs

generally lead to better performance but increase computational time. Our tuning explored values of 8, 16, and 32, with 16 emerging as the optimal value.

- **Learning rate**: controls how much each tree contributes to the final prediction. Smaller values set mean smaller contributions from each tree. Our tuning explored a range from 0.01 to 0.30, identifying an optimal value of 0.186.
- **Tree depth**: controls the complexity of each individual tree. Higher values allow the model to capture more complex patterns but risk overfitting. Our tuning explored depths from 3 to 14, with a depth of 5 showing the best performance.
- **Number of estimators**: represents the total number of trees in the ensemble. More trees typically lead to better performance but with diminishing returns and increased computational cost. Our tuning explored a range of 50-300 trees, with 161 as optimal.
- **Subsample ratio**: represents the proportion of samples used for training each tree. By training each tree on different subsets of data, this parameter helps prevent overfitting. Our tuning explored values from 0.5 to 1.0, with 0.551 proving optimal.
- **Column sampling by tree**: is similar to subsample ratio but for features. This parameter randomly selects different features for each new tree. Our tuning explored values from 0.5 to 1.0, with a relatively high value of 0.832 emerging as optimal, suggesting that the majority of features are valuable for our ranking model.
- **Minimum child weight**: is a regularization technique that controls the threshold for creating additional splits. Our tuning explored values from 1 to 9, with a low value of 2 providing the best performance without overfiting.

Appendix E. Supporting results

Table 10: Log-rank test for differences between sponsored and organic listing

Comparison	Test Statistic	p-value
Hourly survival	2379.63	0.0
Daily survival	272.49	3.26×10^{-61}

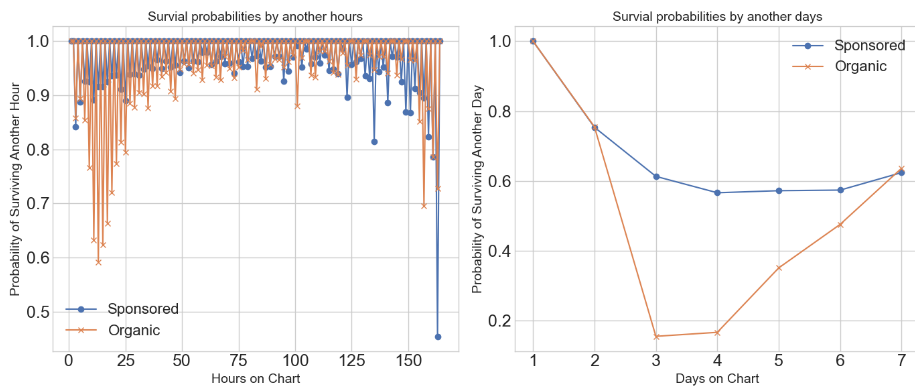

Fig. 7: The probability of products surviving another hour or day.

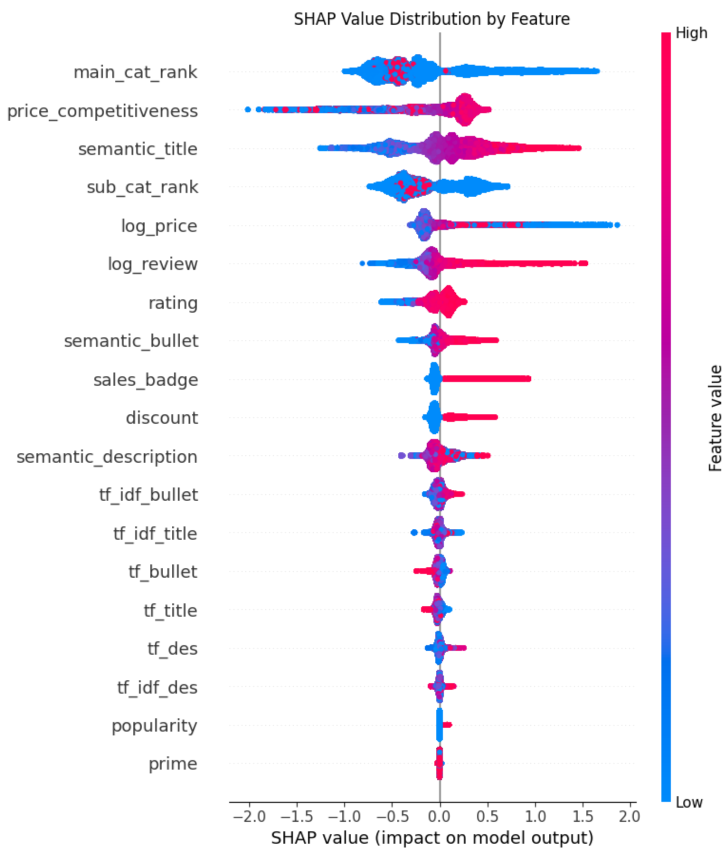

Fig. 8: SHAP summary

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Lists of Abbreviations

Abbreviation	Description
ACoS	Advertising Cost of Sales
API	Application Programming Interface
ASIN	Amazon Standard Identification Number
BERT	Bidirectional Encoder Representations from Transformers
BSR	Best Seller Rank
CDP	Chrome DevTools Protocol
CR	Conversion Rate
CTR	Click-Through Rate
GBT	Gradient Boosted Trees
HTML	HyperText Markup Language
IDF	Inverse Document Frequency
KM	Kaplan-Meier
KV	Keyword
LambdaMART	Lambda Multiple Additive Regression Trees
LSI	Latent Semantic Indexing
LTR	Learning to Rank
nDCG	Normalized Discounted Cumulative Gain
NLP	Natural Language Processing
PHM	Proportional Hazards Model
PPC	Pay-Per-Click
S-BERT	Sentence-BERT
SEA	Search Engine Advertising
SEO	Search Engine Optimization
SERP	Search Engine Results Page
SHAP	SHapley Additive exPlanations
TACoS	Total Advertising Cost of Sales
TF	Term Frequency
TF-IDF	Term Frequency–Inverse Document Frequency
XGBoost	eXtreme Gradient Boosting

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